

JNTU Kakinada

MID-TERM EXAMINATION ? 2025

Control Systems ? Mid-Term Examination, 2025

Subject Code: EC403 Semester: 4 Regulation: R23

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Transfer Functions and Block Diagrams)

1. Discuss the engineering applications of Transfer Functions and Block Diagrams. [12 Marks]
2. Write short notes on Transfer Functions and Block Diagrams. [12 Marks]
3. Write short notes on Transfer Functions and Block Diagrams. [12 Marks]

SECTION B (Unit 2: Time Response Analysis)

4. Solve a problem related to Time Response Analysis. Show all the steps clearly. [12 Marks]
5. Write short notes on Time Response Analysis. [12 Marks]
6. Write short notes on Time Response Analysis. [12 Marks]

SECTION C (Unit 3: Root Locus Technique)

7. Define Root Locus Technique. Explain its significance in Control Systems with a suitable example. [8 Marks]
8. Discuss the engineering applications of Root Locus Technique. [12 Marks]
9. How is Root Locus Technique implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION D (Unit 4: Frequency Response and Stability)

10. Write short notes on Frequency Response and Stability. [12 Marks]
11. Describe the working principle of Frequency Response and Stability with necessary equations. [12 Marks]
12. Derive the expression related to Frequency Response and Stability and explain each term. [12 Marks]

SECTION E (Unit 5: State Space Analysis)

13. List the advantages and limitations of State Space Analysis. Explain any two in detail. [8 Marks]
14. List the advantages and limitations of State Space Analysis. Explain any two in detail. [8 Marks]
15. Define State Space Analysis. Explain its significance in Control Systems with a suitable example. [12 Marks]

*** End of Question Paper ***